

CS 483 Project 4

Virtual Dice Roller

Objective

In this project, you will create a simple interactive web application using HTML, CSS, and JavaScript. The goal is to build a webpage where users can roll a virtual die, and the outcome is displayed visually. I will provide all necessary assets, including dice images and a background image, as well as a semantic HTML structure that you must use. Additionally, the styling should be done according to the provided visual guide. Your task is to style the page with CSS and implement the interactivity with JavaScript.

Project Requirements and Guideline

HTML:

- Use the provided HTML structure, without any alterations, as the foundation for your webpage. It includes the necessary elements such as:
 - `<header>` with a heading `<h1>` and a subtitle `<h2>`
 - A `<section>` containing the Roll the Dice button and a `<figure>` for the dice image
 - A `<footer>` displaying the last rolled timestamp
 - The `<figure>` and `<figcaption>` tags are used to associate the dice image with its description

CSS:

- Use an external CSS file
- Background:
 - The background image should be set in CSS using the `background-image` class in HTML
 - Look into the following CSS properties: `background-image`, `background-size`, `background-position`, `background-repeat`, `position`, `z-index`, etc.
- Button Styling:
 - The button should have a hover effect where the background changes and the cursor is set to a pointer to indicate interactivity
- Dice Image Styling:
 - The image should be hidden by default (using `visibility: hidden`) and only become visible when a dice roll occurs
 - You may want to set a fixed width of 100px with automatic height to preserve the aspect ratio

- Footer Styling:
 - The footer should display the "Last rolled" message as indicated in the visual guide
- Responsive Design:
 - Use a media query for screens with a maximum width of 768px
 - Adjust the content layout to make the page more user-friendly on smaller devices, such as reducing the margin and adjusting the font sizes for the headings, button, etc.

JavaScript:

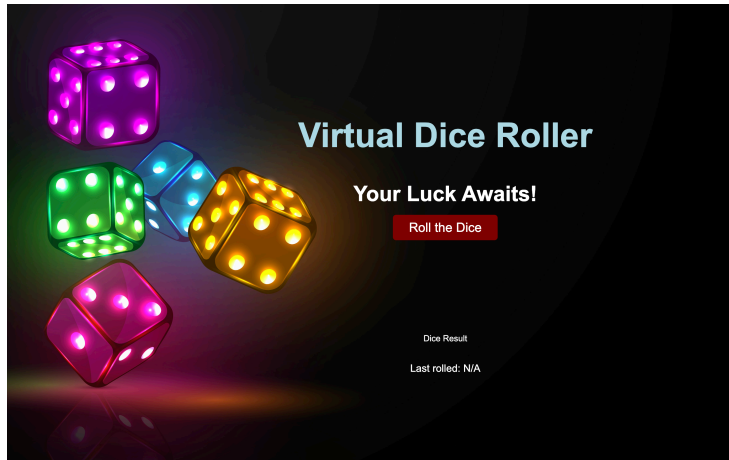
- Use an external JavaScript file
- Implement functionality to randomly select a dice face when the user clicks the roll button
- Dynamically update the dice image to reflect the outcome of the dice roll
- Dynamically update the last rolled date and time
- Include console logging for debugging purposes to track the outcome of each roll

Provided Assets

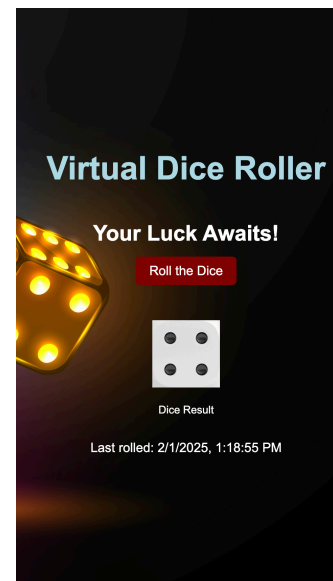
You will receive the following assets to complete the project:

- Dice images (dice1.png to dice6.png)
- Background image (dice_bg.jpeg)

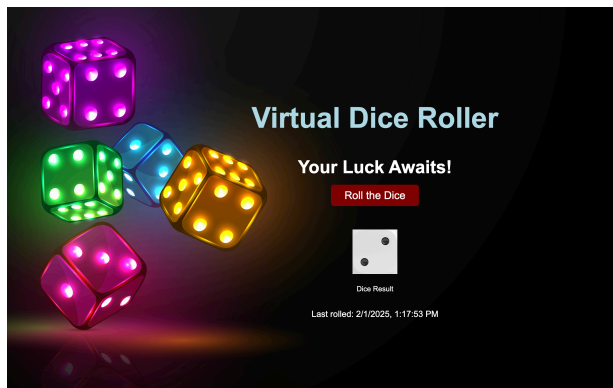
Visual Guide and Design Reference



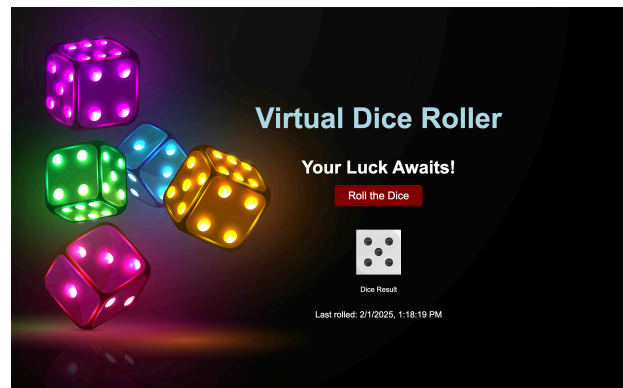
Initial View: Webpage loaded with the 'Roll the Dice' button, ready for interaction



Mobile View of the App



After Dice Roll: Outcome of the dice roll is displayed, along with the updated timestamp showing the time of the last roll



View after another dice roll

Score Breakdown

Criteria	Point %
JavaScript Functionality: ▪ Proper Event Handling (15%): "Roll the Dice" button triggers the appropriate event to generate a random dice roll ▪ Random Dice Roll (15%): Generates a random number between 1 and 6 and dynamically updates the displayed dice image accordingly ▪ Dynamic Update of Timestamp (15%): Displays the date and time of the last dice roll ▪ Console logging (5%): Logs the result of each dice roll	50%
CSS Styling and Responsiveness: ▪ Proper Background Setup (10%) ▪ Adherence to Visual Style Guide (10%): Content is centered, spaced appropriately, and not visually overlapping with the background image ▪ Button Styling with Clear Visual Clues (5%) ▪ Responsiveness (5%)	30%
Provided semantic HTML structure is used without modification	15%
Clean Code Style, Documentation, README	5%

Submitting your Solution

Organize your HTML, CSS and JavaScript files into the correct directory structure, ensuring that the grader can render the HTML without any extra steps. Once organized, compress the files into a ZIP archive and name it as follows: **<first name>_<last name>_proj4.zip**. Note that requiring additional communication from the grader to view or test your application will result in deductions.

Your project archival should include a text file named **README** with the following information:

- Your **name** and **email address**
- A brief proposal for a **future enhancement** of your application

Good luck, and have fun rolling the dice!